Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the test accuracy during training.

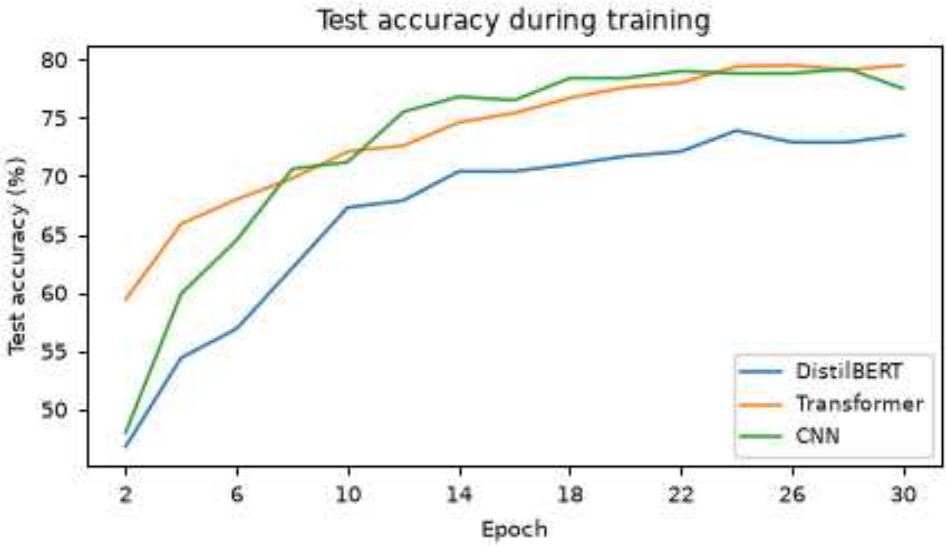


Figure 1: Test accuracy during training for the models in this experiment.

From epoch 18 to epoch 30, the test accuracy of Transformer improves by a relative 6.0%. From epoch 12 to epoch 22, the test accuracy of CNN improves by a relative 5%. At epoch 26, Transformer is 6.6 points better than DistilBERT. As shown in Figure 1, from epoch 20 to epoch 26, the test accuracy of DistilBERT improves by a relative 1.7%.

Figure 2 shows the test accuracy during training.

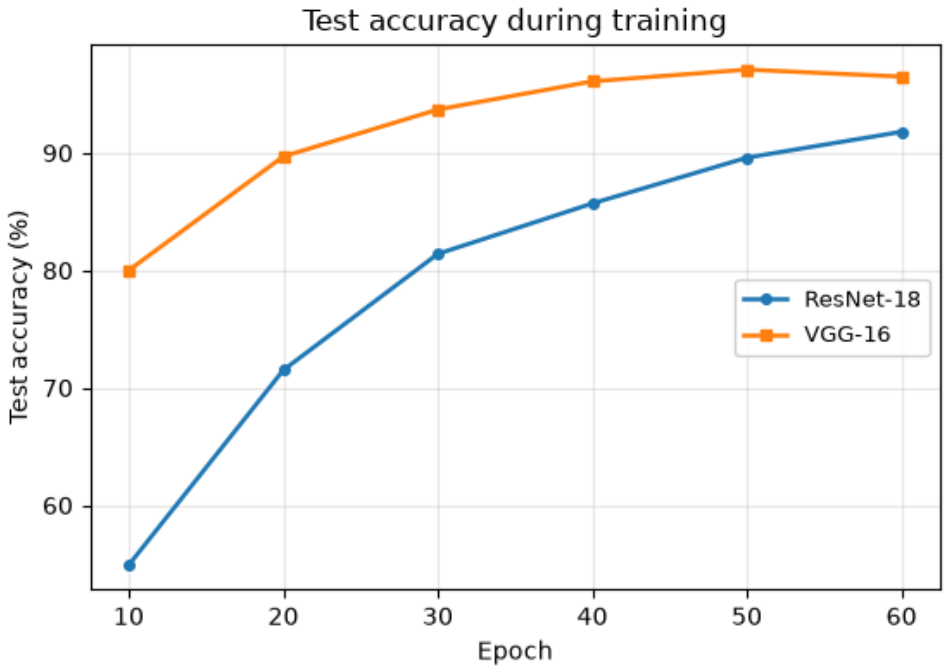


Figure 2: Test accuracy during training for the models in this experiment.

As shown in Figure 2, at epoch 60, VGG-16 is 3.8 points better than ResNet-18. As shown in Figure 2, ResNet-18 reaches over 70% test accuracy at epoch 20.

Figure 3 shows the f1 score during training.

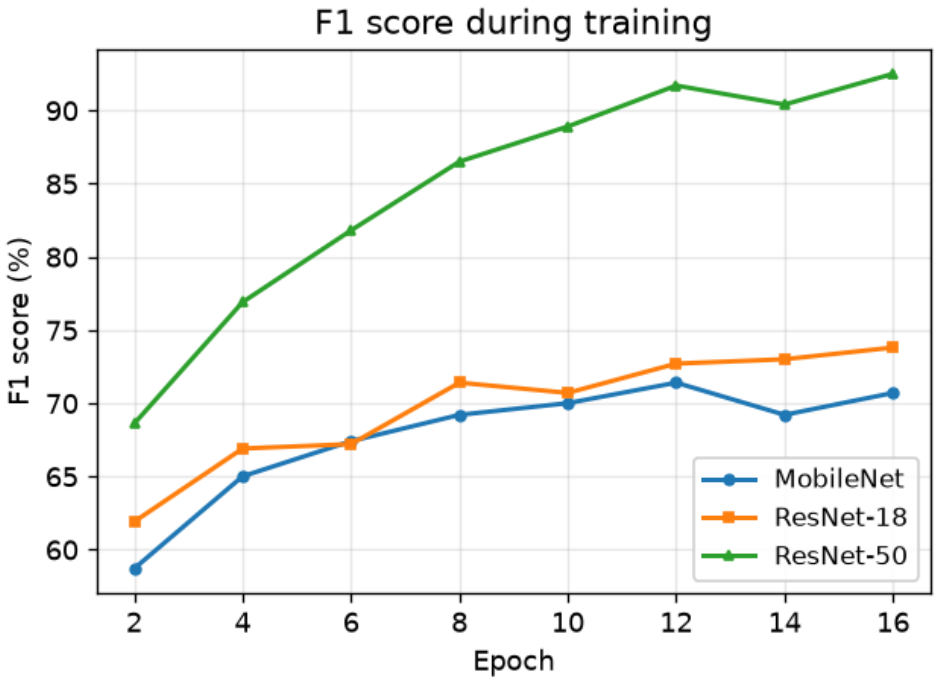


Figure 3: F1 score during training for the models in this experiment.

As shown in Figure 3, ResNet-50 outperforms ResNet-18 by 19 percentage points at epoch 16. At epoch 8, ResNet-18 achieves 67.7% F1 score. As shown in Figure 3, ResNet-18 reaches 73.0% F1 score at epoch 14.

3 Discussion

On CIFAR-10, SVM obtains an accuracy of 83.6%. We note that the training was stable for all models.